Timing

File: timing.kicad_sch

Sprite Table

File: sprite_table.kicad_sch

Sprite RAM

Sprite ROM 1

File: sprite_rom_1.kicad_sch

Sprite ROM logic

PWB(C)

Sprite ROM 2

File: sprite_rom_logic.kicad_sch File: sprite_rom_2.kicad_sch

PWB(C) is used when sprite ROMs P13,P16,P15 and P18 on PWB(B) are not mounted.

File: sprite_ram.kicad_sch

Sprite Frame Buffers

File: sprite_frame_buffers.kicad_sch

Spare parts on video PWB(B) 302191A

B_B9C 74LS04

B_B9B 74LS04

B_B9E 74LS04

B_B9D 74LS04

B_P11B 74LS74

B_R11A 74LS32

Spare parts are set as NC.

Spare parts on main PWB(A) 302190A

A_L2A 74LS00

A_L2D 74LS00

A_H8A 74LS07

A_H8D 74LS07

A_K1C 74LS32

A_L1C 74LS07

A_L1B 74LS07

A_L8F 74HC367

A_N3B 74LS139

GND

KiCad does not support reference designators starting with a number. Therefore the references used in the original schematics have been mirrored. 6R -> R6. The original schematics had separate pages for the main and video PCB. They are combined here. This makes some references duplicate. To distinguish between boards A_ is added before part positions on PWB(A) and for PWB(B) B_ and so on. If there are still conflicts a "P" (Position) is added after board letter AP_, BP. Sometimes there are duplicate reference designators then "CAP" maybe added after board letter ACAP_ (CAP is for capacitor).

Tile Layers

File: tile_layers.kicad_sch

IO

File: io.kicad_sch

Color mixer

File: color_mixer.kicad_sch

Sound

File: sound.kicad_sch

Sound mixer

File: sound_mixer.kicad_sch

Timing diagrams

File: timing_diagrams.kicad_sch

Sprite timing diagrams

File: sprite_timing_diagrams.kicad_sch

Main CPU

File: mcpu.kicad_sch

Shared Resources

File: shared_resources.kicad_sch

Slave CPU

File: scpu.kicad_sch

Fixed Layer

File: fixed_layer.kicad_sch

Only used on GX903 – Cue Brick

Battery Powered RAM

File: bpram.kicad_sch

There are several games and variations on the twin 16 platform:
- Main Board PWB(A)
- Video Board PWB(B)
- Additional ROM Board, if not ROM's mounted directly on video board.
- Battery powered RAM board

	Id	PWB(A)	PWB(B)	PWB(C)
Dark Adventure	GX687	302190A	302191A	–
Gradius 2	GX785	302190A	302191A	–
Vulcan Venture	GX785	302190A	302191A	–
Gradius 2	GX785	302190B	302191B	402145A
Hard Puncher	GX870	302190B	302191B	402145A
M.I.A	GX808	302190B	302191B	402145A
Cue Brick	GX903	302190B	302191B	–

Cue Brick has an additional Battery Powered RAM Board: PWB451803

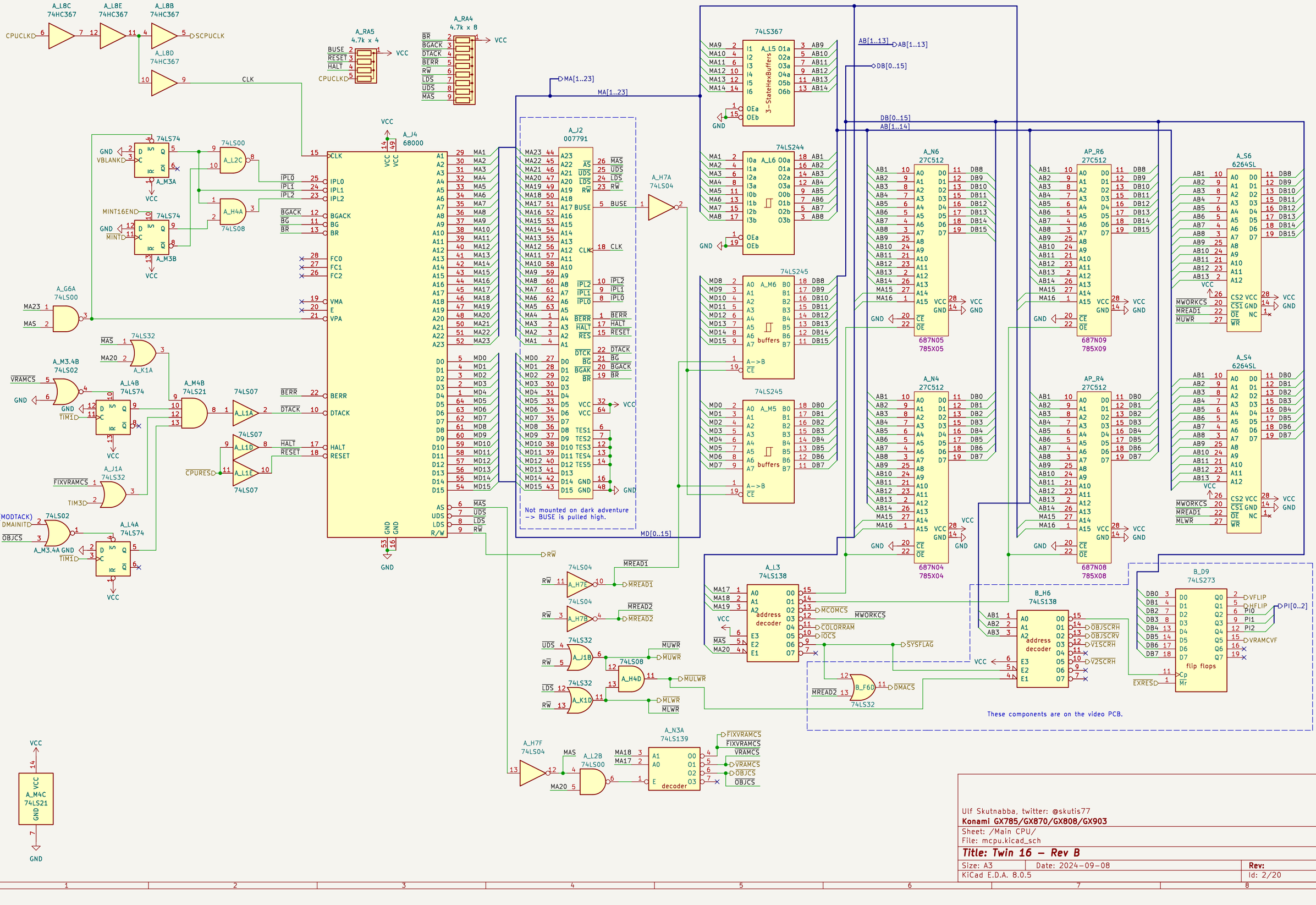
Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /
File: twin16b.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 1/20


A_L8C 74HC367
A_L8E 74HC367
A_L8B 74HC367
A_L8D 74HC367
CPUCLKD
SCPUCLK
CLK
A_RA5 4.7k x 4
BUSE
RESET
HALT
CPUCLKD
VCC
A_RA4 4.7k x 8
BR
BGACK
DTACK
BERR
RW
LDS
UDS
MAS
VCC
MA[1..23]
A_J4 68000
A_J2 007791
Not mounted on dark adventure -> BUSE is pulled high.
MD[0..15]
74LS367
3-StateHexBuffers
AB[1..13]
DB[0..15]
74LS244
74LS245
buffers
A->B
DB[0..15]
AB[1..14]
A_N6 27C512
687N05 785X05
AP_R6 27C512
687N09 785X09
A_S6 6264SL
A_N4 27C512
687N04 785X04
AP_R4 27C512
687N08 785X08
A_S4 6264SL
MWORKCS
MREAD1
MLWR
MREAD2
MUWR
74LS74
A_M3A
A_M3B
74LS00
A_L2C
A_H4A
74LS08
IPL0
IPL1
IPL2
BGACK
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
74LS07
A_L1A
A_L1D
A_L1E
CPURES
A_G6A 74LS00
A_K1A 74LS32
A_M3.4B 74LS02
A_L4B 74LS74
A_M4B 74LS21
A_J1A 74LS32
A_L4A 74LS74
A_M4C 74LS21
A_H7A 74LS04
A_H7E 74LS04
A_H7B 74LS04
A_J1B 74LS32
A_K1D 74LS32
A_H4D 74LS08
A_H7F 74LS04
A_L2B 74LS00
A_N3A 74LS139
A_L3 74LS138
address decoder
B_H6 74LS138
B_F6D 74LS32
B_D9 74LS273
flip flops
These components are on the video PCB.
Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903
Sheet: /Main CPU/
File: mcpu.kicad_sch
Title: Twin 16 – Rev B
Size: A3
Date: 2024-09-08
Rev:
KiCad E.D.A. 8.0.5
Id: 2/20

Only used on PWB(A)302190B
Goes to CN2:24.
On PWB(A)302190A CN2:24 is connected to GND.

In the original schematics
DMAINT, MODTACK and DMAINIT are the same

8R and 10R are not mounted when
the GX687 board is used for gradius 2
with GX785 roms.

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Slave CPU/
File: scpu.kicad_sch

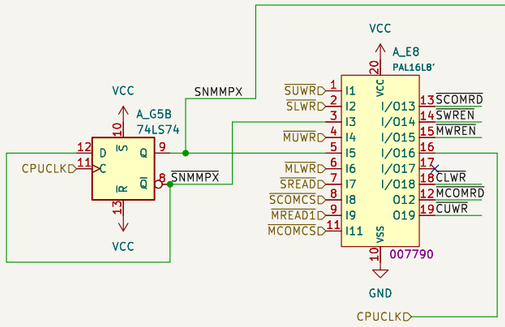
Title: Twin 16 – Rev B

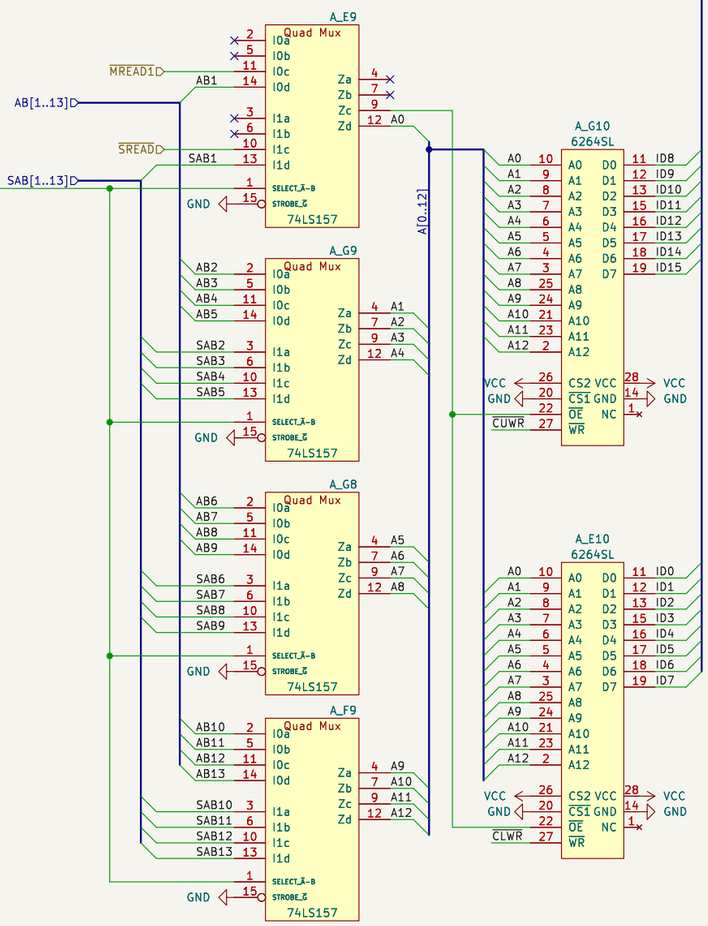
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 3/20

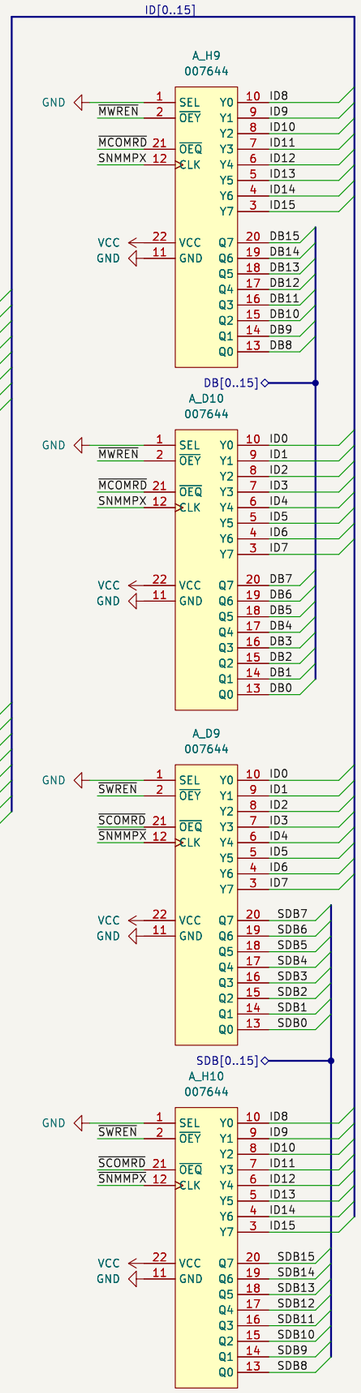
Konami 007790

$\overline{MCOMRD}$ = ~(~$\overline{MREAD1}$ & ~$\overline{MCOMCS}$) = $\overline{MREAD1}$ | $\overline{MCOMCS}$
$\overline{SCOMRD}$ = ~(~$\overline{SREAD}$ & ~$\overline{SCOMCS}$)
$\overline{SWREN}$ = ~(~SNMMPX & $\overline{SREAD}$ & ~$\overline{SCOMCS}$)
$\overline{MWREN}$ = ~(~SNMMPX & $\overline{MREAD1}$ & ~$\overline{MCOMCS}$)
$\overline{CLWR}$ = ~(~$\overline{SLWR}$ & ~$\overline{SNMMPX}$ & SNMMPX & $\overline{SREAD}$ & ~$\overline{SCOMCS}$ & ~CPUCLK | ~SNMMPX & ~$\overline{MLWR}$ & $\overline{MREAD1}$ & ~$\overline{MCOMCS}$ & ~CPUCLK)
$\overline{CUWR}$ = ~(~$\overline{SUWR}$ & ~$\overline{SNMMPX}$ & SNMMPX & $\overline{SREAD}$ & ~$\overline{SCOMCS}$ & ~CPUCLK | ~$\overline{MUWR}$ & ~SNMMPX & $\overline{MREAD1}$ & ~$\overline{MCOMCS}$ & ~CPUCLK)

All AND logic can be converted to OR form. But since AND logic and inverters are used in the PAL16L8 device it is kept in that form.
The equations follow verilog syntax. The active low signals are not inverted, the bar is there only to show activeness.







This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs
mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs.
If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.

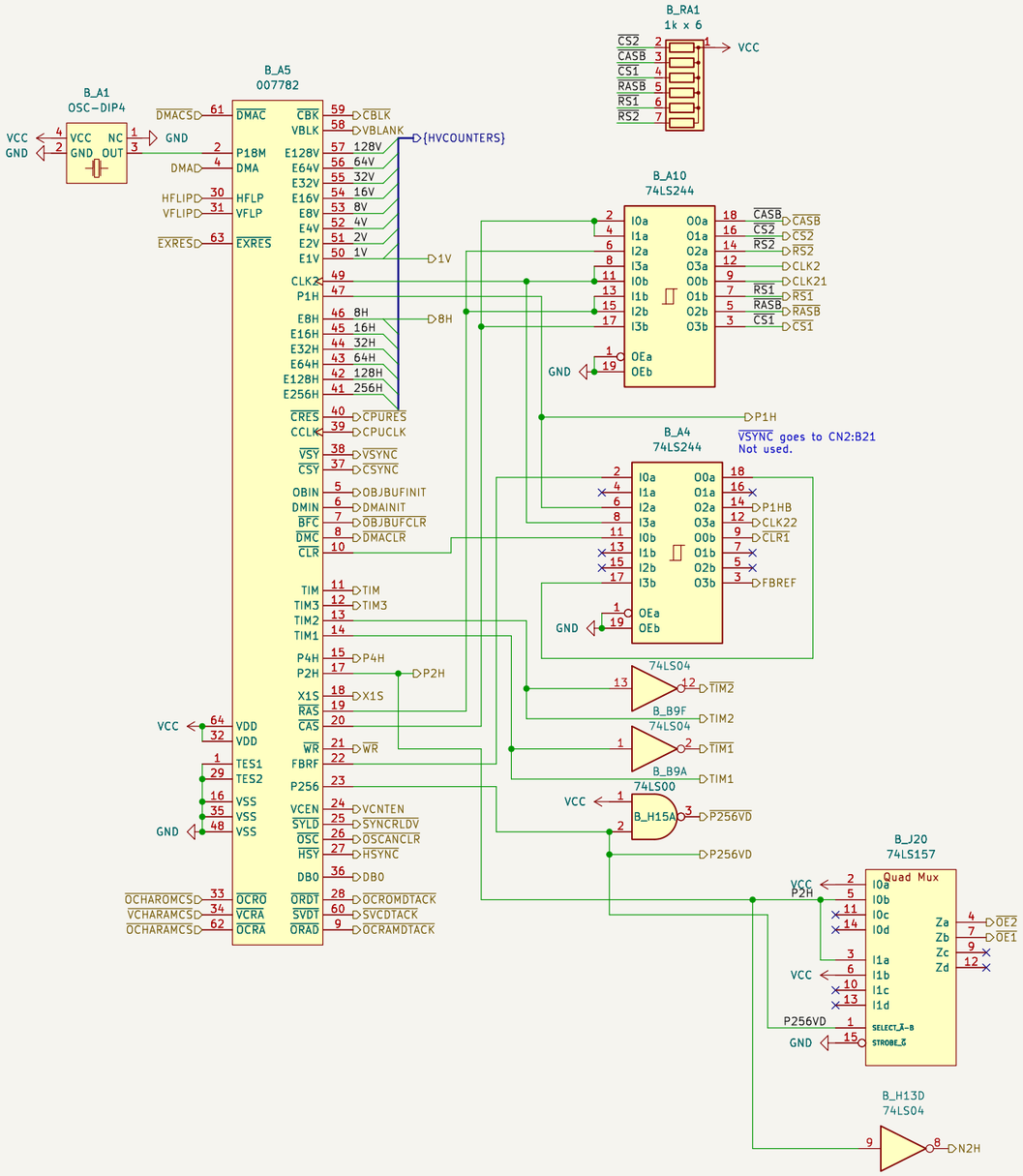
Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

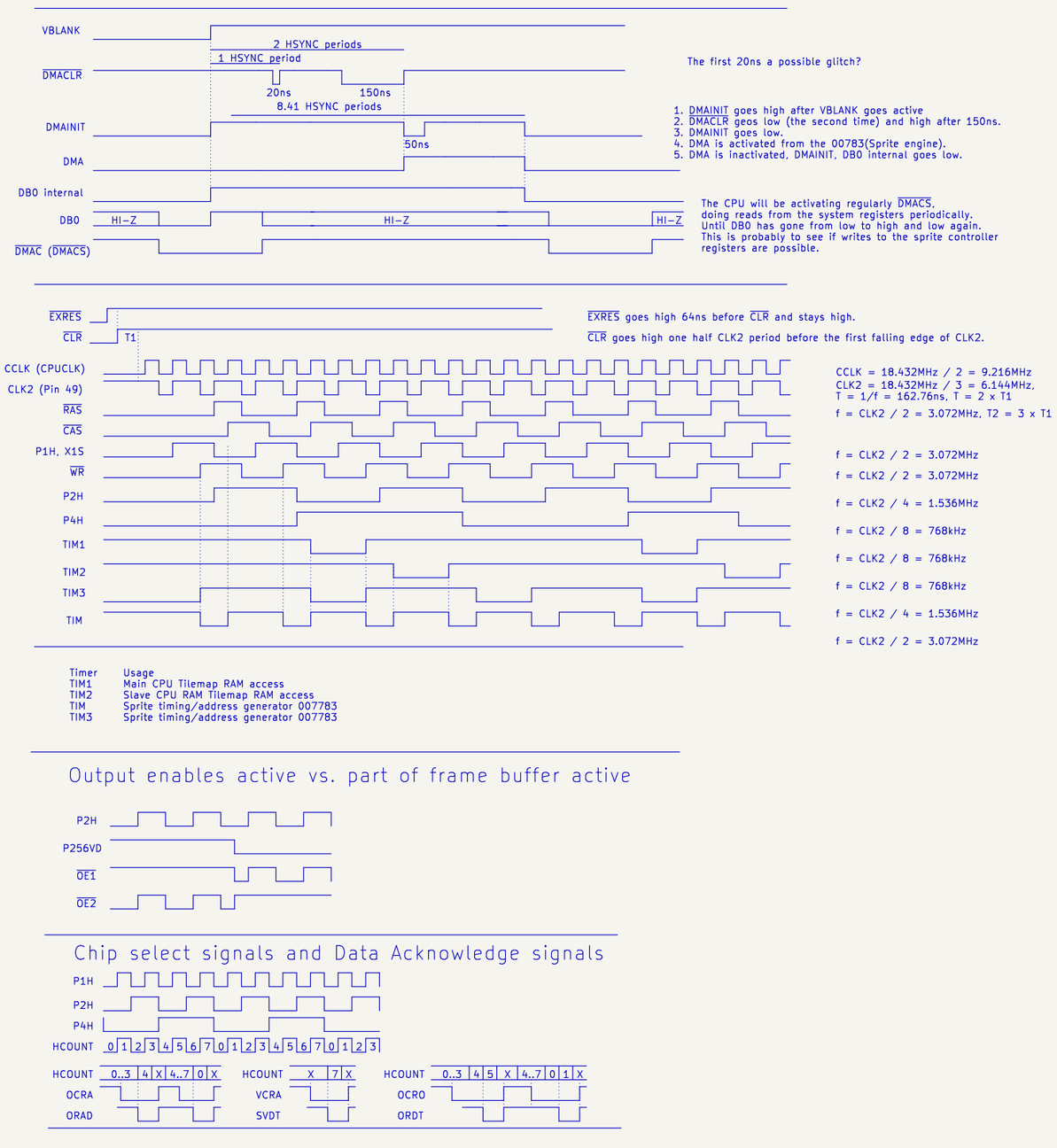
Sheet: /Shared Resources/
File: shared_resources.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 4/20



DMA Synchronization



Timer	Usage
TIM1	Main CPU Tilemap RAM access
TIM2	Slave CPU RAM Tilemap RAM access
TIM	Sprite timing/address generator 007783
TIM3	Sprite timing/address generator 007783

Output enables active vs. part of frame buffer active

Chip select signals and Data Acknowledge signals

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Timing/
File: timing.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 5/20

B_B3
74LS157
Quad Mux
AB[1..13]
AB12
AB13
GND
P2HD
SELECT_A-B
STROBE_E
A11
A12
B_B4
74LS157
AB8
AB9
AB10
AB11
16V
32V
64V
128V
A7
A8
A9
A10
B_B5
74LS157
AB4
AB5
AB6
AB7
64H
128H
256H
8V
A3
A4
A5
A6
A[0..12]
B_E4A
74LS27
B_H13F
74LS04
FIXVRAMCS
MLWRD
B_B6
74LS157
MREAD2D
AB1
AB2
AB3
8H
16H
32H
OE
A0
A1
A2
10ns delay
B_E4B
74LS27
B_H13A
74LS04
MUWRD
B_C3
6264SL
B_C4
6264SL
D[0..15]
CS2
CS1
OE
WR
VCC
GND
NC
7ns delay
B_A11A
74LS86
P4HD
P4H
20ns delay
B_D5
74LS174
B_C6
74LS174
B_D6
74LS174
D9
D10
D11
D12
D13
D14
D15
D0
D1
D2
D3
D4
D5
D6
D7
D8
hex flip flop
Cp
Mr
1V
2V
4V
50ns delay from OE active
to output data valid
B_F7
74LS245
B_F8
74LS245
DB15
DB14
DB13
DB12
DB11
DB10
DB9
DB8
DB7
DB6
DB5
DB4
DB3
DB2
DB1
DB0
buffers
A->B
CE
B_F6A
74LS32
{HVCOUNTERS}
DB[0..15]
RAM
A[0..5] Horizontal position 0..504 in steps of 8
A[6..10] Vertical position 0..248 in steps of 8
A[11.12] Tilemaps? Up to 4.
D[0..8]: Tile number
D[9..12]: Palette
D13: HFLIP
D14: VFLIP
Palette, HFLIP and VFLIP is valid for
eight pixels at a time and becomes valid
on the rising edge of P4H
7ns delay
B_A11C
74LS86
P1HBD
P1HB
B_A11B
B_A11C
B_B11A
B_B11B
B_B11C
B_B11D
B_B10C
B_B10B
B_B10D
HFLIP
B_D8
27C128
785G14
870F14
808E14
903E14
GND
VCC
CE
OE
PGM
VPP
ROM
A[0..1] Horizontal position 0..6 in steps of 2
A[2..4] Vertical position 0..7
A[5..13] Tile number
Two pixels color data is output
on each rising edge of P1HB.
The current pixel data is multiplexed
with P1HB (and HFLIP).
B_B7
74LS273
FX[20..23]
FX20
FX21
FX22
FX23
B_B8
74LS157
FX[10..13]
FX10
FX11
FX12
FX13
B_A11D
74LS86
B_B10A
The total delay for a pixel is seven pixel clocks
from when P2H and OE goes high and low respectively.
B_C9
74LS273
B_C10
74LS273
B_D10
74LS273
flip flops
CLK21
FX0
FX1
FX2
FX3
FX4
FX5
FX6
FX7
FX[0..7]
FX[0..3]: Color
FX[4..7]: Palette
3. Data out latched
into DFF's.
4. DFF's data
out valid
6. DFF's data
out valid
CLK21
P1H
P1HB
P2H
P4H
2. Data out from
RAM valid
5. Palette and color values
latched into DFF's 3C and 2E.
4. FX[0..7] Data out valid
1. Addresses and OE
latched into RAM. This condition sets the start:
P2H & P4H = 1.

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Fixed Layer/
File: fixed_layer.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 6/20

This is from jammarcade.net

Konami 007644

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.

The 007781 is a shift register. All counters related to all timing signals are inside the chip. It is synchronized on the rising edge of CLR and CLK2 after that. It reads 32 bits / 4 bits / pixel = 8 pixels at a time.

1C[0:3]: Layer 1 Pixel color code
1C[4:6]: Layer 1 Palette
2C[0:3]: Layer 2 Pixel color code
2C[4:6]: Layer 2 Palette

VD[0..12]: Tile number
VD[13..15]: Palette Color

Data latched from the RAM:
Edge Tilemap
T1–T2 1
T2–T3 2

It is output 2 pixel clocks later to the ROMs.

When VA is switched to output the bus is updated for each P1H cycle.

The vertical counters are not the same value when they are offset with the scroll registers.

Tilemap 1 and 2
VA[5..0]: Horizontal position: 0..63 -> 0..63 x 8 = 0..504
VA[11..6]: Vertical position: 0..63 -> 0..63 x 8 = 0..504
VA12 = 0: Tilemap 1, VA12 = 1: Tilemap 2

SYNCRLDV: Usually 60.604Hz
T is ~150ns.

VCNTEN: 8.0000kHz

HSYNC: 16.0000kHz, T2: 5.2175us

Outputs of RAM active when P4H is low, i.e during T1 and T2.

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Tile Layers/
File: tile_layers.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 7/20

B_B12
007783

B_C13
6264SL

B_C14
6264SL

B_C12
007784

B_F12
007785

74LS157
Quad Mux
B_J13

74LS157
Quad Mux
B_J14

74LS157
Quad Mux
B_J18

74LS157
Quad Mux
B_J16

74LS157
Quad Mux
B_J17

74LS245
B_F10

74LS245
B_F9

74LS245
B_J8

74LS245
B_J9

B_F6B
74LS32

B_H12C
74LS32

B_H12A
74LS32

B_K12A
74LS32

B_P11A
74LS74

B_A16
PAL16L8'

B_RA3
2.2k x 8

B_RA2
2.2k x 8

AB[1..13]
SAB[1..14]
OA[0..12]
CRW[0..3]
OD[0..15]
DB[0..15]
SDB[0..15]
SA[15..20]
O[0..19]
SPA[0..19]
VD[7..0]
HD[8..2]
HJ[0..1]
CX[0..3]
C[0..3]
{RW1}
{RW2}

On PWB(A) 302190A CHAPAGE
(CN2:A23) is connected to GND

Each unparsed sprite (object) occupies 40 words
in object RAM. Only the first 8 words of each
object data is used. The memory range for
unparsed data is up 17FFh. That leaves room
for 153 (153 x 40 −1 = 17E7h) objects per frame.

007784 Sprite parser

− subtracts the scroll value for the
sprites from current coordinates for the sprites.
− When DMA goes high. The 007784 will read 8 words, if bit OD15
of the first word is high then BEFLAG goes high and stays high
until the sprite RAM has processed the first four words and
possibly updated the four following words at the high end of
the RAM. It will compress the eigth words into four
words and also subtract the x− and y−scroll values from the
x and y coordinates. It will write the four words at the high
end of the sprite RAM.
− Addresses at the low end goes from 0x0000..0x0007 and the
following cycle at 0x0028..0x002F..

Scroll Register
OBJSCRV Sprite horizontal scroll
OBJSCRH Sprite vertical scroll

Data is latched in the scroll registers when
the signals OBJSCRV and OBJSCRH go low.

007793

CLK21′ = CLK21

LRW = -(DMAONS & OBJBUFINIT & CLK21′ |
BEFLAG & DMA & P1H & DMAONS & CLK21′ & (~8H & HFLIP | 8H & ~HFLIP)
~LWR & ~DMAINIT)

URW = -(DMAONS & OBJBUFINIT & CLK21′ |
BEFLAG & DMA & P1H & DMAONS & CLK21′ & (8H & ~HFLIP | ~8H & HFLIP) |
~UWR & ~DMAINIT)

The sprite memory is controlled in three separate cases:
1. OBJBUFINIT
2. BEFLAG & DMA
3. ~DMAINIT

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Sprite Table/
File: sprite_table.kicad_sch

Title: Twin 16 − Rev B

Size: A3	Date: 2024−09−08	Rev:
KiCad E.D.A. 8.0.5		Id: 8/20

Graphics ROM and RAM

ROM naming

	ID	P13	P15	P16	P18
cuebrickj	GX903	NP	NP	NP	NP
miaj	GX808	808D15	NP	808D17	NP
hpuncher	GX870	870C15	870C16	870C17	870C18
gradius2 / vulcan	GX785	785F15	785F16	785F17	785F18

NP = Not Populated
On some boards P13,P15,P16 and P18 are NP and replaced with PWB(C) 402145A. See separate schematics.

This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs. If SEL is LOW then the outputs will only change state on a rising edge CLK.

Sprite Character Banks

Page	SPA19	SPA18	SPA17
1	X	L	L
2	X	L	H
3	L	H	L
4	H	H	H
5 RAM	X	H	H

Pixel color data output rate is:
16-bits / 2 pixels = 8 bits / pixel.

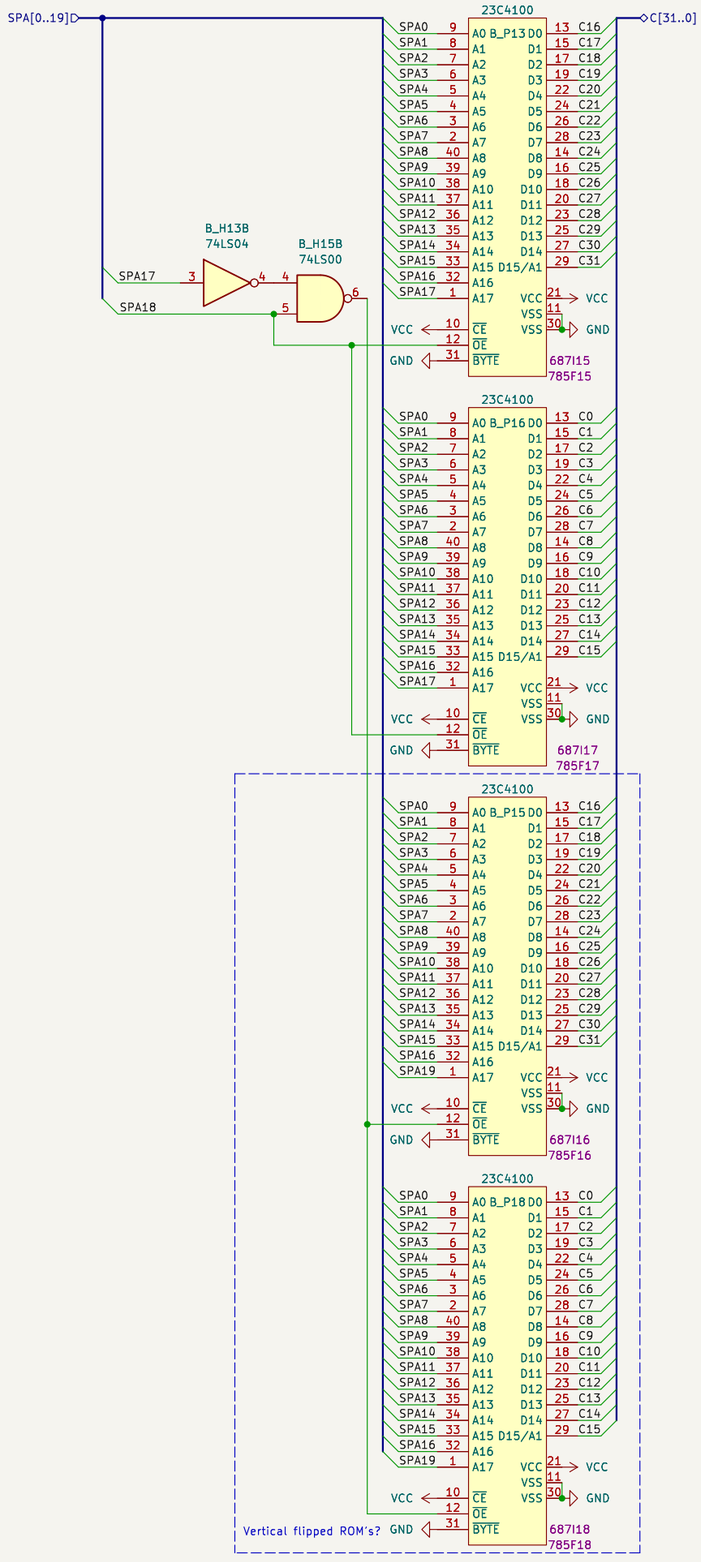
W[15]: Vertical flip

Konami 007788

The inverted ~P2H is written as N2H.

$\overline{OCRDL}$ = ~(~$\overline{SREAD}$ & SAB1 & ~$\overline{OCHARAMCS}$ & $\overline{OCHAROMCS}$ | ~$\overline{SREAD}$ & SAB1 & ~$\overline{OCHAROMCS}$)
$\overline{OCWL}$ = ~(~P2H & $\overline{SREAD}$ & SAB1 & ~$\overline{OCHARAMCS}$)
$\overline{OCRDH}$ = ~(~$\overline{SREAD}$ & ~SAB1 & ~$\overline{OCHARAMCS}$ & $\overline{OCHAROMCS}$ | ~$\overline{SREAD}$ & ~SAB1 & ~$\overline{OCHAROMCS}$)
$\overline{OCWH}$ = ~(~P2H & $\overline{SREAD}$ & ~SAB1 & ~$\overline{OCHARAMCS}$)
$\overline{VCWENL}$ = ~(P2H & P4H & ~$\overline{VCHARAMCS}$ & $\overline{SREAD}$ & SAB1)
$\overline{VCRDL}$ = ~(~$\overline{VCHARAMCS}$ & ~$\overline{SREAD}$ & SAB1)
$\overline{VCRDH}$ = ~(~$\overline{VCHARAMCS}$ & ~$\overline{SREAD}$ & ~SAB1)
$\overline{VCWENH}$ = ~(P2H & P4H & ~$\overline{VCHARAMCS}$ & $\overline{SREAD}$ & ~SAB1)
$\overline{P2HP4H}$ = ~(P2H & P4H)
$\overline{N2HP4H}$ = ~(~P2H & P4H)

Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903
Sheet: /Sprite RAM/
File: sprite_ram.kicad_sch
Title: Twin 16 – Rev B
Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 8.0.5 | Id: 9/20



Sheet: /Sprite ROM 1/	
File: sprite_rom_1.kicad_sch	
Title: Twin 16 – Rev B	
Size: A3	Date: 2024-09-08
KiCad E.D.A. 8.0.5	Id: 10/20

C_A14
4.7k x 8
C'0 C'1 C'2 C'3 C'4 C'5 C'6 C'7
VCC

C'[31..0]
C[31..0]

C_A15
C_D11
C_C11
C_B15
74LS244
VCC
GND

C_C12A
74LS08
SPA17
SPA18

SPA[19..0]

CR_D8
4.7k x 8
SPA7 SPA6 SPA5 SPA4 SPA3 SPA2 SPA1 SPA0

C_D9
C_C9
SPA0 SPA1 SPA2 SPA3 SPA4 SPA5 SPA6 SPA7
SPA8 SPA9 SPA10 SPA11 SPA12 SPA13 SPA14 SPA15
SPA'0 SPA'1 SPA'2 SPA'3 SPA'4 SPA'5 SPA'6 SPA'7
SPA'8 SPA'9 SPA'10 SPA'11 SPA'12 SPA'13 SPA'14 SPA'15
SPA'[15..0]

C_C8
4.7k x 8
SPA9 SPA10 SPA8 SPA11 SPA12 SPA15 SPA13 SPA14

C_D10
C_C10
4.7k x 8
C'8 C'9 C'10 C'11 C'12 C'13 C'14 C'15
C'16 C'17 C'18 C'19 C'20 C'21 C'22 C'23

C_B14
4.7k x 8
C'24 C'25 C'26 C'27 C'28 C'29 C'30 C'31

Mirrored when connected to PWB(B)

C_CN3
DIN64PS
SPA9 SPA8 SPA7 SPA6 SPA5 SPA4 SPA3 SPA2 SPA1 SPA0
SPA10 SPA11 SPA12 SPA13 SPA14 SPA15 SPA16 SPA17 SPA18 SPA19
C15 C14 C13 C12 C11 C10 C9 C8 C7 C6 C5 C4 C3 C2 C1 C0
C31 C30 C29 C28 C27 C26 C25 C24 C23 C22 C21 C20 C19 C18 C17 C16

C_D12
4.7k x 4
SPA16 SPA17 SPA18

C_D13
74LS138
address decoder
SPA16 SPA17 SPA18
SPOE0 SPOE1 SPOE2 SPOE3

SPOE[7..0]

C_C15B
C_C15A
C_C15C
C_C15D
74LS32
SPOE4 SPOE5 SPOE7 SPOE6

C_C13A
74LS04
SPA19

C_C12B
C_C12C
C_C12D
74LS08

C_C13B
C_C13C
C_C13D
C_C13E
C_C13F
74LS04

PWB(C) 402145A
Konami GX785/GX870/GX808/GX903
Sheet: /Sprite ROM logic/
File: sprite_rom_logic.kicad_sch
Title: Twin 16 – Rev B
Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 8.0.5 | Id: 11/20

C_B12 C_B11 C_B10 C_B9 C_A12 C_A11 C_A10 C_A9
C_B8 C_B7 C_B6 C_B5 C_B4 C_B3 C_B2 C_B1
C_D8 C_D7 C_D6 C_D5 C_D4 C_D3 C_D2 C_D1
C_A8 C_A7 C_A6 C_A5 C_A4 C_A3 C_A2 C_A1
27C512
F17B F17D F17F F17H F18B F18D F18F F18H
F17A F17C F17E F17G F18A F18C F18E F18G
F15B F15D F15F F15H F16B F16D F16F F16H
F15A F15C F15E F15G F16A F16C F16E F16G
SPA'0 SPA'1 SPA'2 SPA'3 SPA'4 SPA'5 SPA'6 SPA'7 SPA'8 SPA'9 SPA'10 SPA'11 SPA'12 SPA'13 SPA'14 SPA'15
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15
D0 D1 D2 D3 D4 D5 D6 D7
C'0 C'1 C'2 C'3 C'4 C'5 C'6 C'7 C'8 C'9 C'10 C'11 C'12 C'13 C'14 C'15 C'16 C'17 C'18 C'19 C'20 C'21 C'22 C'23 C'24 C'25 C'26 C'27 C'28 C'29 C'30 C'31
GND VCC
CE OE
SPOE0 SPOE1 SPOE2 SPOE3 SPOE4 SPOE5 SPOE6 SPOE7
On the PCB the position is D9
SPOE[7..0]
SPA'[15..0]
C'[31..0]

Konami GX785/GX870/GX808/GX903	
Sheet: /Sprite ROM 2/	
File: sprite_rom_2.kicad_sch	
Title: Twin 16 – Rev B	
Size: A3	Date: 2024-09-08
KiCad E.D.A. 8.0.5	Id: 11/20

Frame buffer 1

Framebuffer 1 is outputting pixel data when P256VD is low and writes to the FB is possible when P256VD is high. The conditions are reveresed for FB2.

Not certain which of the parts 6KB,6UB and 6UD are used where. Some errors in the original schematics.

Frame buffer 2

Writes to FB are possible when the "pixel part" is sleected from the mux.
If this expression is true
(R[0:3] != 4'h0) && (FBWM1 | FBWM2 | -WR | -VPD8)
data is written to DRAM.

Data is written when:
– WR is active.
– Not a transparent pixel (4'h0).
– FBWM1 and FBWM2 are not active: No writes to frame buffer going on.
– VPD8 (vertical counter highest bit, VCNT > 256) is not active. Outside visible area.

OCD[0:3]: Pixel color
OCD[4:7]: Palette

Frame buffer 1 Palette color

Frame buffer 2 Palette color

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Sprite Frame Buffers/
File: sprite_frame_buffers.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 13/20

Components are on Video PCB

Components are on Main PCB

the four bits of each layer has the following meaning:
0000: Transparent
1111: Shadow
Else: Opaque

Konami 007789

i11 (1C6) and i5 (PI2) is not used

~o15 = OCD0 & OCD1 & OCD2 & OCD3 | ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3

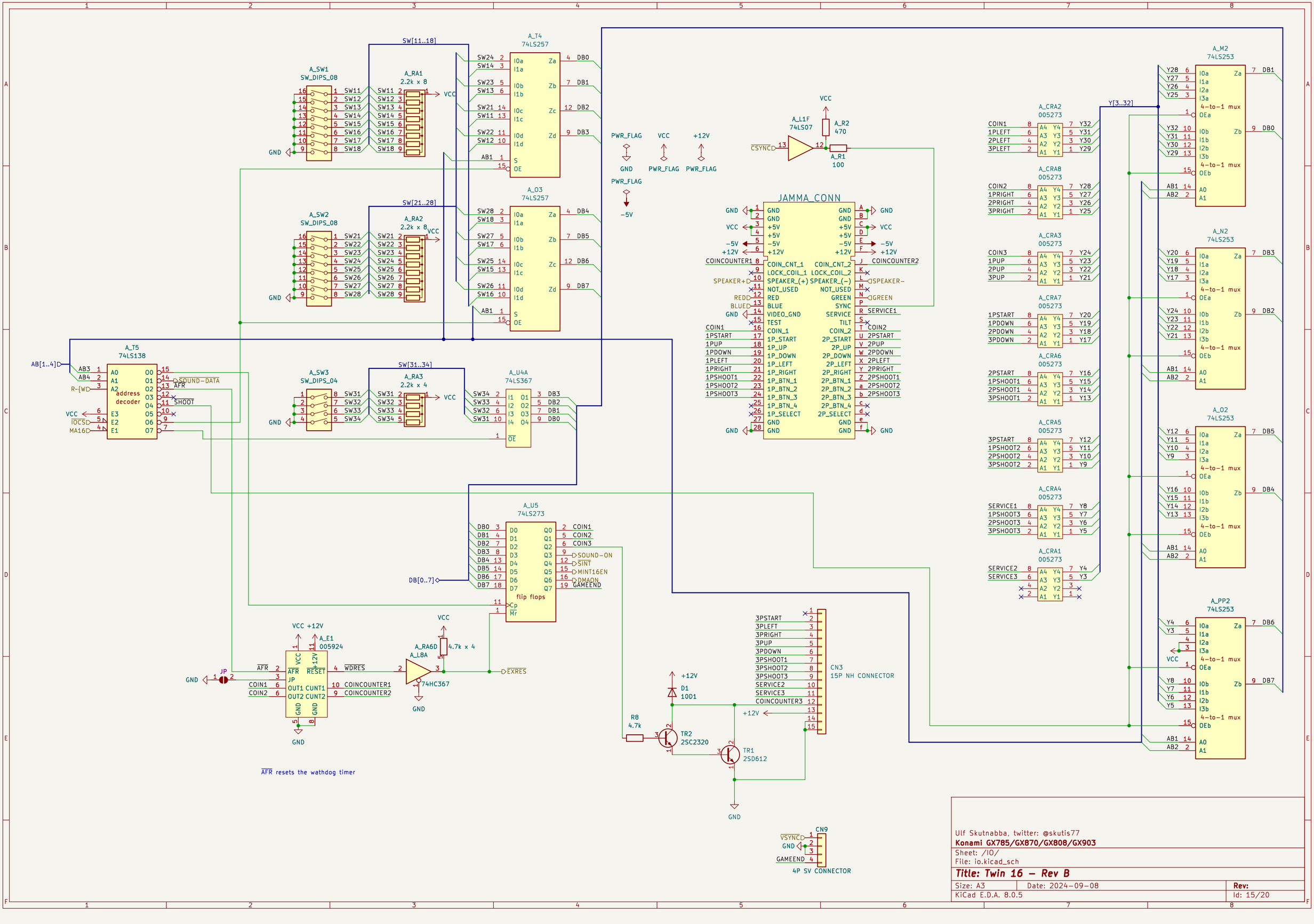
o16 = OCD0 & OCD1 & OCD2 & OCD3

SELA = 1C |
PI0 & ~PI1 & FX |
~PI1 & OCD0 & OCD1 & OCD2 & OCD3 & FX |
~PI1 & ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3 & FX |
PI1 & ~2C0 & OCD0 & OCD1 & OCD2 & OCD3 & ~2C1 & ~2C2 & ~2C3 |
PI1 & ~2C0 & ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3 & ~2C1 & ~2C2 & ~2C3

~SELB = ~1C & OCD0 & OCD1 & OCD2 & OCD3 |
~1C & ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3 |
~1C & PI0 & ~PI1 & FX |
~1C & PI0 & PI1 & 2C0 & BUS |
~1C & PI0 & PI1 & BUS & 2C1 |
~1C & PI0 & PI1 & BUS & 2C2 |
~1C & PI0 & PI1 & BUS & 2C3

~SHADOW = ~1C & ~PI1 & OCD0 & OCD1 & OCD2 & OCD3 |
~1C & ~PI0 & OCD0 & OCD1 & OCD2 & OCD3 & ~FX |
~1C & PI0 & OCD0 & OCD1 & OCD2 & OCD3 & ~BUS |
~1C & PI0 & ~2C0 & OCD0 & OCD1 & OCD2 & OCD3 & ~2C1 & ~2C2 & ~2C3

Ulf Skutnabba, twitter: @skutis77		
Konami GX785/GX870/GX808/GX903		
Sheet: /Color mixer/		
File: color_mixer.kicad_sch		
Title: Twin 16 – Rev B		
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 14/20



Ulf Skutnabba, twitter: @skutis77

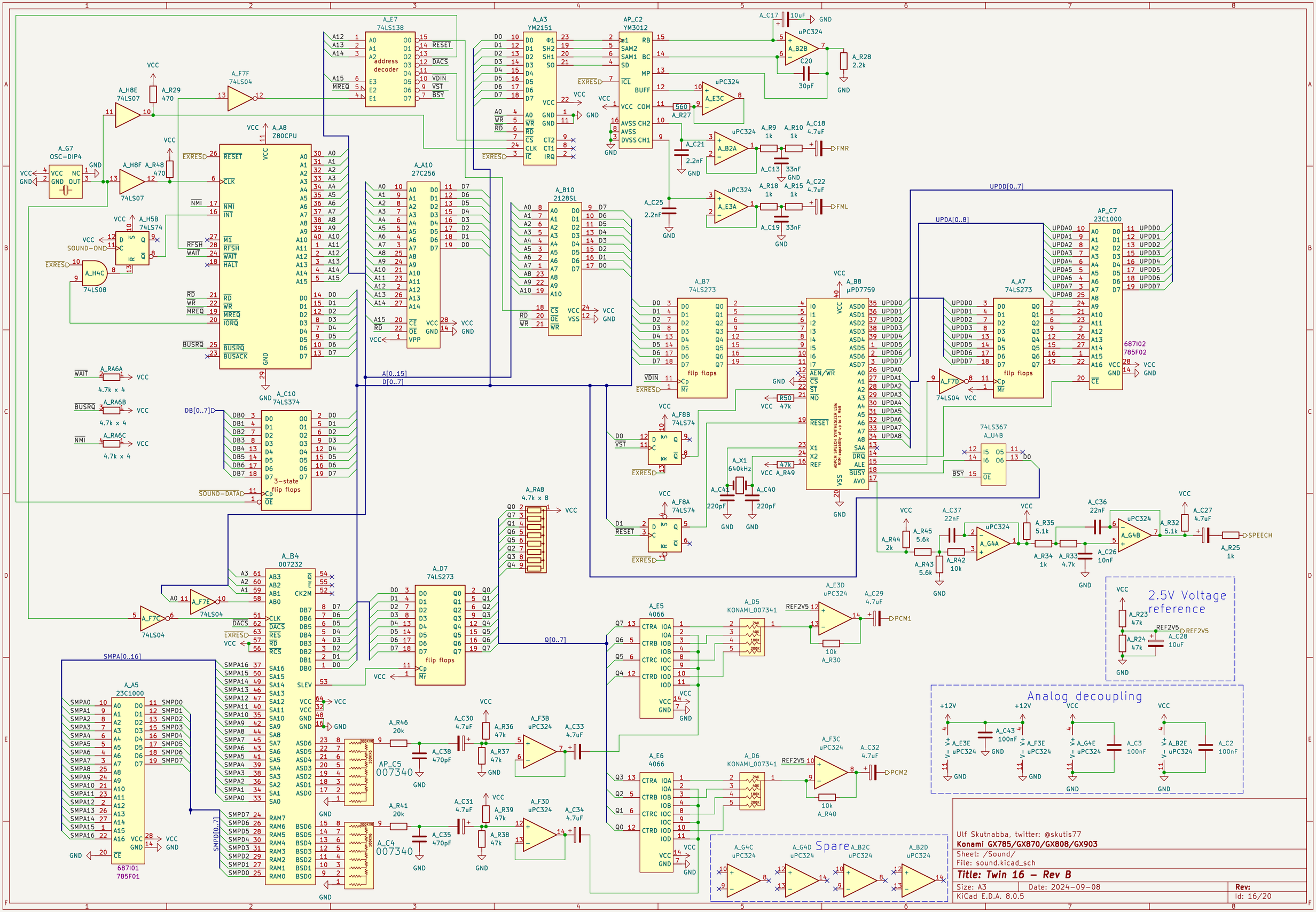
Konami GX785/GX870/GX808/GX903

Sheet: /IO/

File: io.kicad_sch

Title: Twin 16 – Rev B

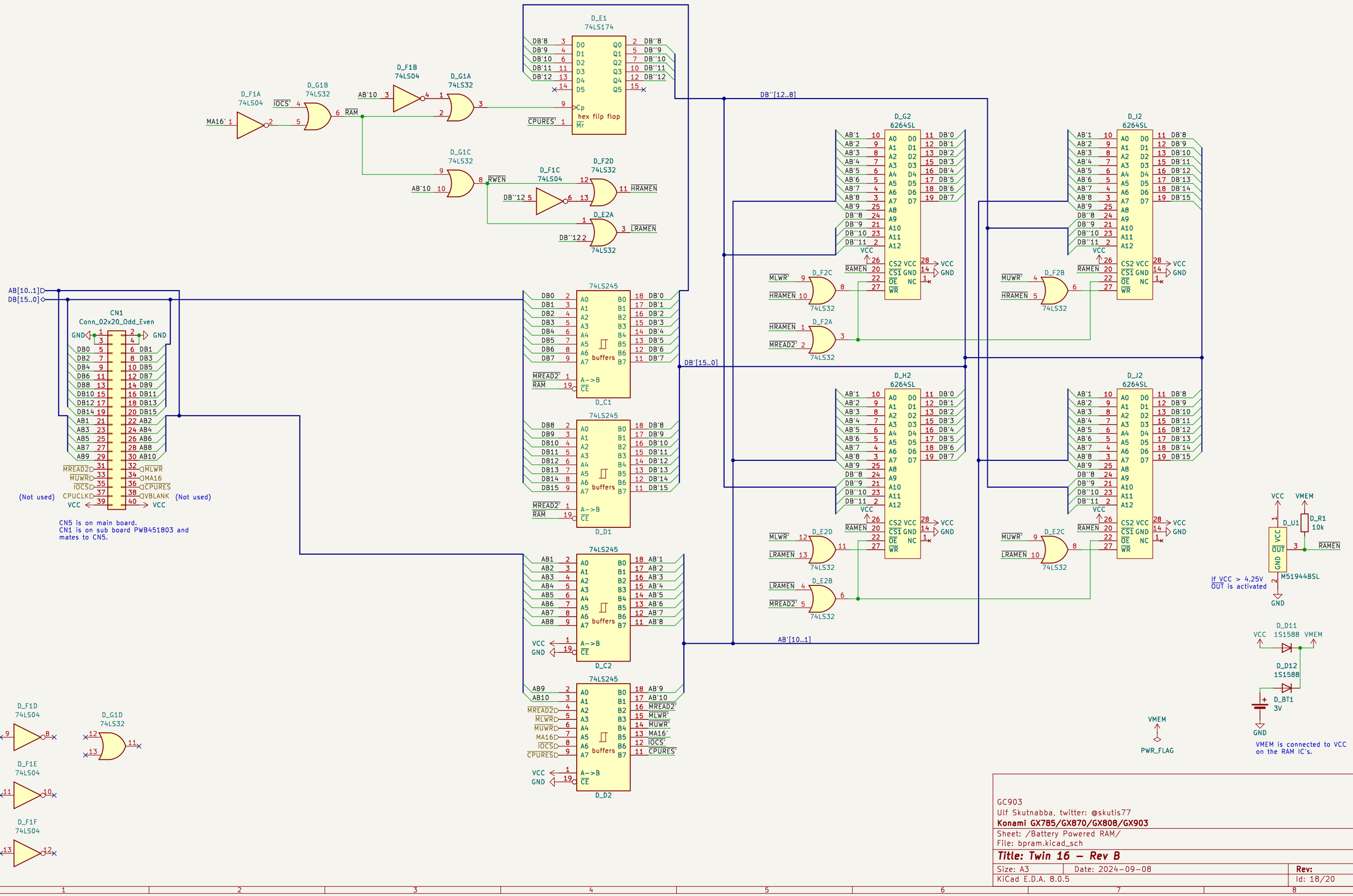
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 15/20


A_E7
74LS138
address decoder
A_A3
YM2151
AP_C2
YM3012
uPC324
A_B2B
A_E3C
A_B2A
A_E3A
A_R28 2.2k
A_C17 10uF
C20 30pF
A_R27 560
A_R9 1k
A_R10 1k
A_C18 4.7uF
A_C13 33nF
A_C21 2.2nF
A_C22 4.7uF
A_C19 33nF
A_C25 2.2nF
FMR
FML
A_G7
OSC-DIP4
A_H8E
74LS07
A_R29 470
A_F7F
74LS04
A_H8F
A_R48 470
A_H5B
74LS74
A_H4C
74LS08
A_A8
Z80CPU
A_A10
27C256
A_B10
2128SL
A_B7
74LS273
A_B8
µPD7759
A_A7
74LS273
AP_C7
23C1000
UPDD[0..7]
UPDA[0..8]
687I02
785F02
A_F7D
74LS04
A_RA6A
A_RA6B
A_RA6C
4.7k x 4
A_C10
74LS374
3-state flip flops
A_F8B
74LS74
A_F8A
74LS74
A_X1
640kHz
A_C41 220pF
A_C40 220pF
A_R49 47k
A_R50 47k
74LS367
A_U4B
A_RA8
4.7k x 8
A_B4
007232
A_D7
74LS273
A_F7E
74LS04
A_F7C
74LS04
A_E5
4066
A_D5
KONAMI_007341
A_E6
4066
A_D6
KONAMI_007341
A_E3D
uPC324
A_C29 4.7uF
A_R30 10k
A_F3C
uPC324
A_C32 4.7uF
A_R40 10k
PCM1
PCM2
A_A5
23C1000
687I01
785F01
AP_C5
007340
A_C4
007340
A_R46 20k
A_R41 20k
A_C38 470pF
A_C35 470pF
A_F3B
uPC324
A_F3D
uPC324
A_C30 4.7uF
A_C31 4.7uF
A_R36 47k
A_R37 47k
A_R39 47k
A_R38 47k
A_C33 4.7uF
A_C34 4.7uF
A_R44 2k
A_R45 5.6k
A_R43 5.6k
A_R42 10k
A_C37 22nF
A_G4A
uPC324
A_R35 5.1k
A_R34 1k
A_R33 4.7k
A_C36 22nF
A_C26 10nF
A_G4B
uPC324
A_R32 5.1k
A_C27 4.7uF
A_R25 1k
SPEECH
2.5V Voltage reference
A_R23 47k
A_R24 47k
A_C28 10uF
REF2V5
Analog decoupling
A_C43 100nF
A_C3 100nF
A_C2 100nF
A_G4C
A_G4D
A_B2C
A_B2D
Spare
A[0..15]
D[0..7]
Q[0..7]
SMPA[0..16]
SMPD[0..7]
DB[0..7]
Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903
Sheet: /Sound/
File: sound.kicad_sch
Title: Twin 16 – Rev B
Size: A3
Date: 2024-09-08
KiCad E.D.A. 8.0.5
Rev:
Id: 16/20

To enable stereo sound output on connector CN8:
Use the connector which shortens Pin 1 -> Pin 2 and
Pin 3 -> Pin 4 on Connector CN7.

A_R17
1k
FMRD
A_R12
7.5k
SPEECHD
A_R11
5.1k
PCM1D
A_R19
5.1k
PCM2D
A_R16
1k
FMLD
A_R22
7.5k
SPEECHD
A_R31
5.1k
PCM1D
A_R20
5.1k
PCM2D
A_E3B
uPC324
REF2V5D
A_R13
12k
+12V
A_R52
3.3k
A_C15
4.7uF
A_R14
10k
A_F3A
uPC324
REF2V5D
A_R21
12k
+12V
A_R53
3.3k
A_C16
4.7uF
A_R26
10k
CN6
4P NH CONN
GND
CN7
4P NH CONN
APOT_R53
1k
APOT_R52
1k
GND
A_C11
100nF
A_C12
100nF
GND
A_C9
100uF
ACAP_C10
100uF
GND
A_D1
MB3722
IN-A
IN+A
IN+A
IN-B
VCC
GNDIN
GNDOUT
BYPASS
BSA
OUTA
BSA
OUTB
+12V
ACAP_C3
1000uF
GND
A_C8
100uF
A_C7
100uF
GND
A_C14
10uF
GND
ACAP_C2
1000uF
A_C6
100nF
A_R6
1
GND
A_C5
100nF
A_R7
1
GND
A_C1
1000uF
SPEAKER-
SPEAKER+
CN8
4P NH CONN
GND

Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903
Sheet: /Sound mixer/
File: sound_mixer.kicad_sch
Title: Twin 16 - Rev B
Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 8.0.5 | Id: 17/20



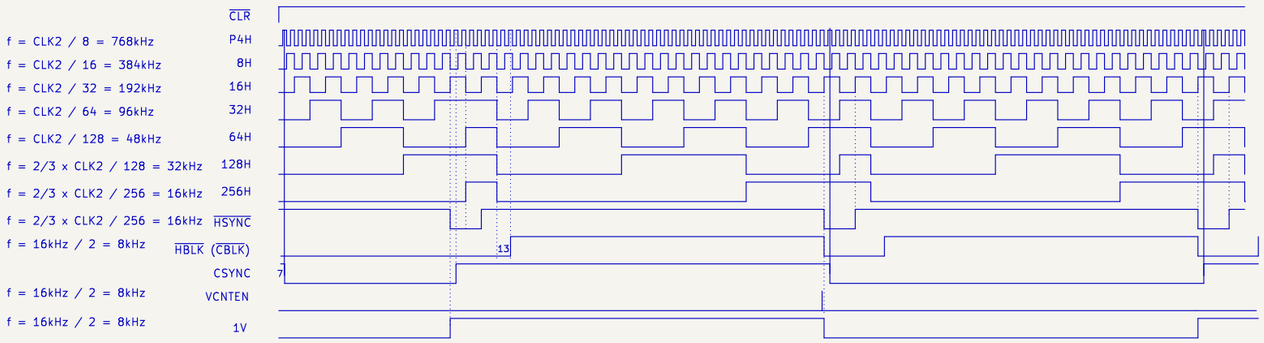
GC903
Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903

Sheet: /Battery Powered RAM/
File: bpram.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 18/20

Horizontal signals

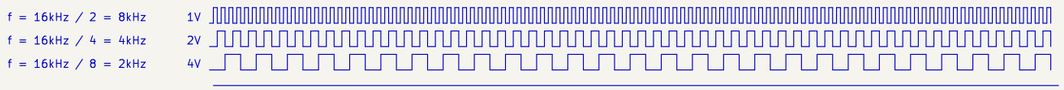


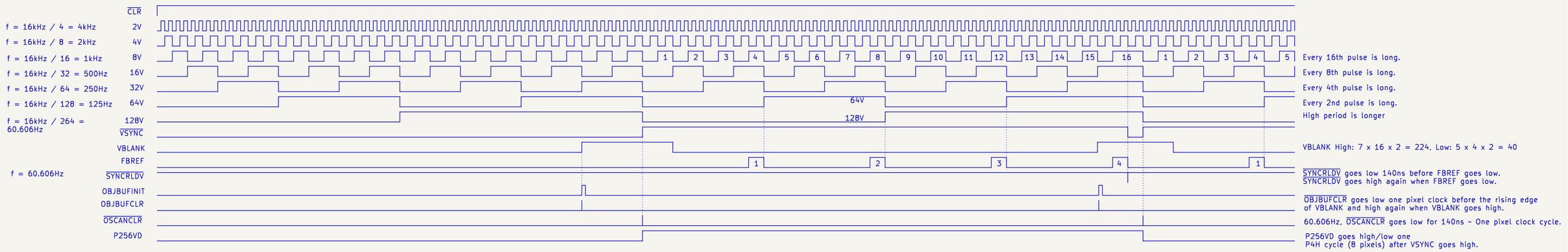
– The first VCNTEN is skipped after reset.
It goes low 140ns before HSYNC goes low,
and high again when HSYNC goes low.
VCNTEN is active right before every second falling edge of HSYNC.

– CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

The first HSYNC is at HCOUNT 175 after reset.

Vertical signals





FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

Horizontal and vertical synch timing diagrams

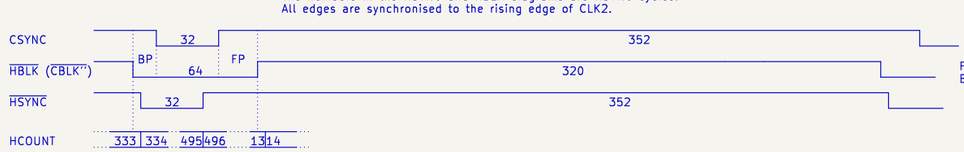
Pixel clock CLK2 = 18.432MHz / 3 = 6.144MHz

Horizontal lines
HSYNC freq = 6.144MHz / 384 = 16kHz
HSYNC period = 1 / 16kHz = 62.5us
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

Vertical lines
VSYNC freq = 16kHz / 264 = 60.606060Hz
VSYNC period = 1 / f = 264 / 16kHz = 16.5ms
scanlines / frame: 264
Active lines: 224
Blank lines: 40

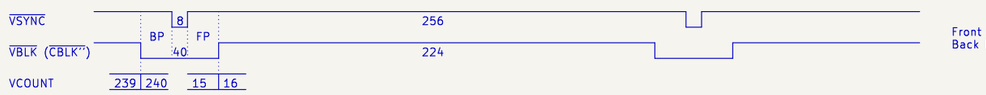
HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

The numbers in the HSYNC and HBLK diagrams are HSYNC cycles.
All edges are synchronised to the rising edge of CLK2.



Front Porch (FP) 24 px,
Back Porch (BP) 8 px

The numbers in the VSYNC and VBLK diagrams are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



Front Porch (FP) and
Back Porch (BP) 16 HSYNC cycles.

CBLK'' is at the output of color mixer 007593.
CBLK' is shifted inside the 007593 one pixel clock.

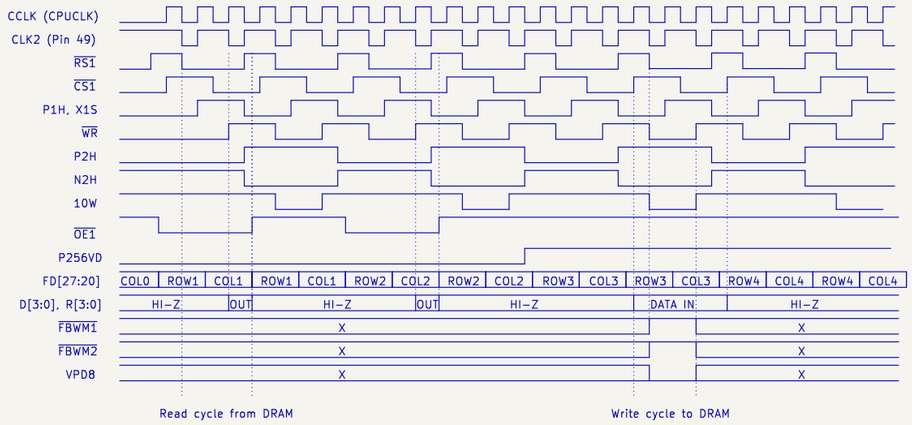
Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Timing diagrams/
File: timing_diagrams.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 19/20



The same row and column addresses are repeated twice.
One for read cycle and the other for write cycles.

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Sprite timing diagrams/
File: sprite_timing_diagrams.kicad_sch

Title: Twin 16 – Rev B

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 20/20